



Design and Implementation of Wi-Fi Jammer using ESP32

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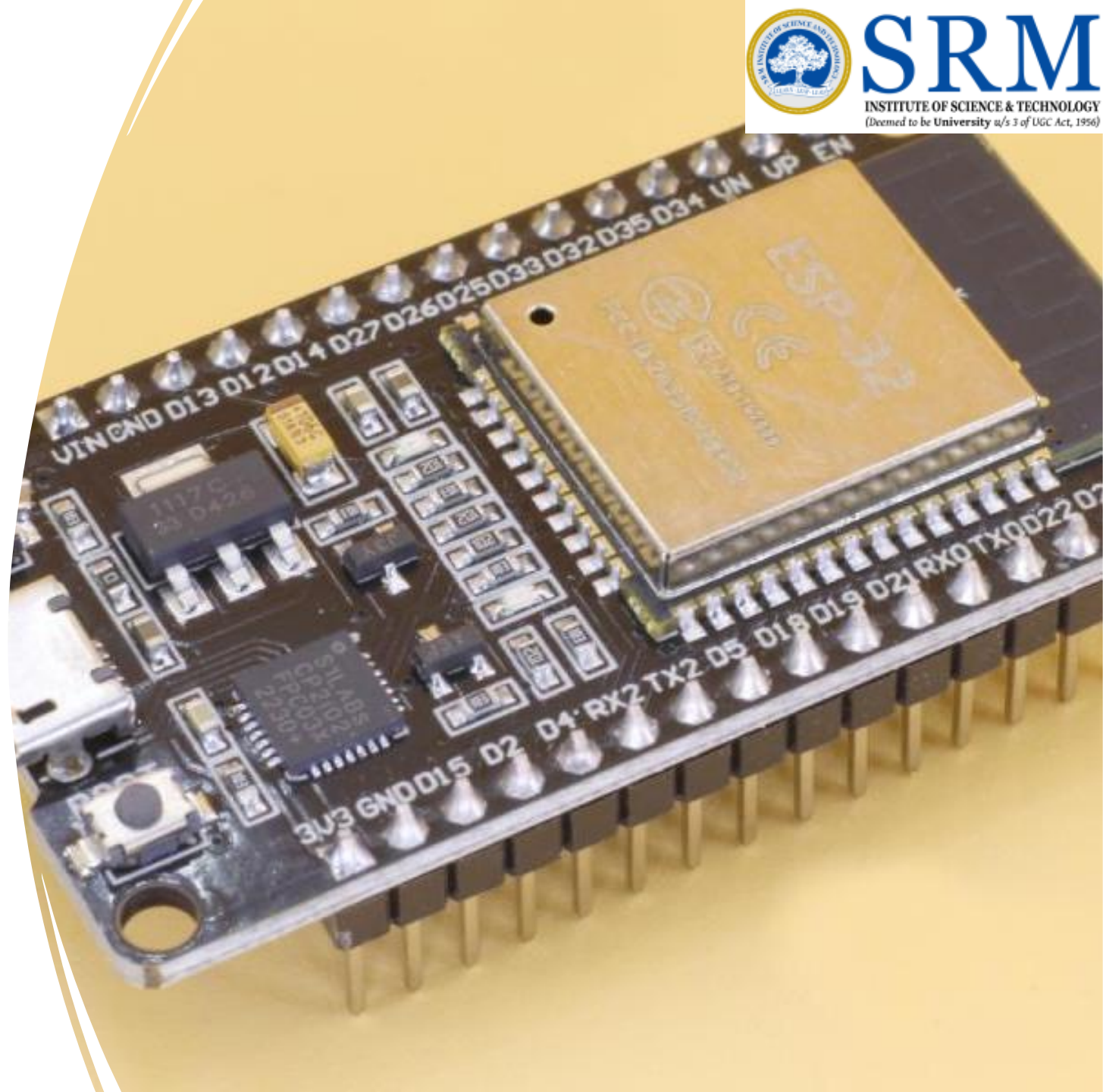
Department: ECE

ABSTRACT

- Design and implementation of a Wi-Fi jammer using ESP32
- Targets vulnerabilities in 2.4 GHz wireless networks
- Scans nearby Wi-Fi networks and captures packets (promiscuous mode)
- Performs deauthentication attacks to disrupt communication
- Displays real-time data (SSID, RSSI, packet count) on OLED
- User control through push buttons
- Low-cost, portable system for security analysis and learning

Domain Identification

- **Domain**
 - Wireless Communication
 - Network Security & Analysis
 - IoT-based Embedded Systems
- **Technology**
 - Wi-Fi (IEEE 802.11)
 - 2.4 GHz ISM Band
 - ESP32 Microcontroller
- **Problem Area**
 - Wi-Fi networks vulnerable to interference
 - Lack of simple tools for packet monitoring
 - Difficulty in analyzing network traffic



Problem Statement

Wi-Fi networks operating in the 2.4 GHz band are prone to interference and packet-based attacks, which can disrupt communication and affect network performance.

Need for Solution

- Analyze network traffic
- Monitor packet transmission
- Demonstrate network vulnerabilities
- Develop low-cost portable system

Objectives

- Design a Wi-Fi jammer using ESP32
- Scan nearby Wi-Fi networks
- Monitor packet transmission using promiscuous mode
- Perform deauthentication attack
- Display network data on OLED screen
- Provide user control using push buttons



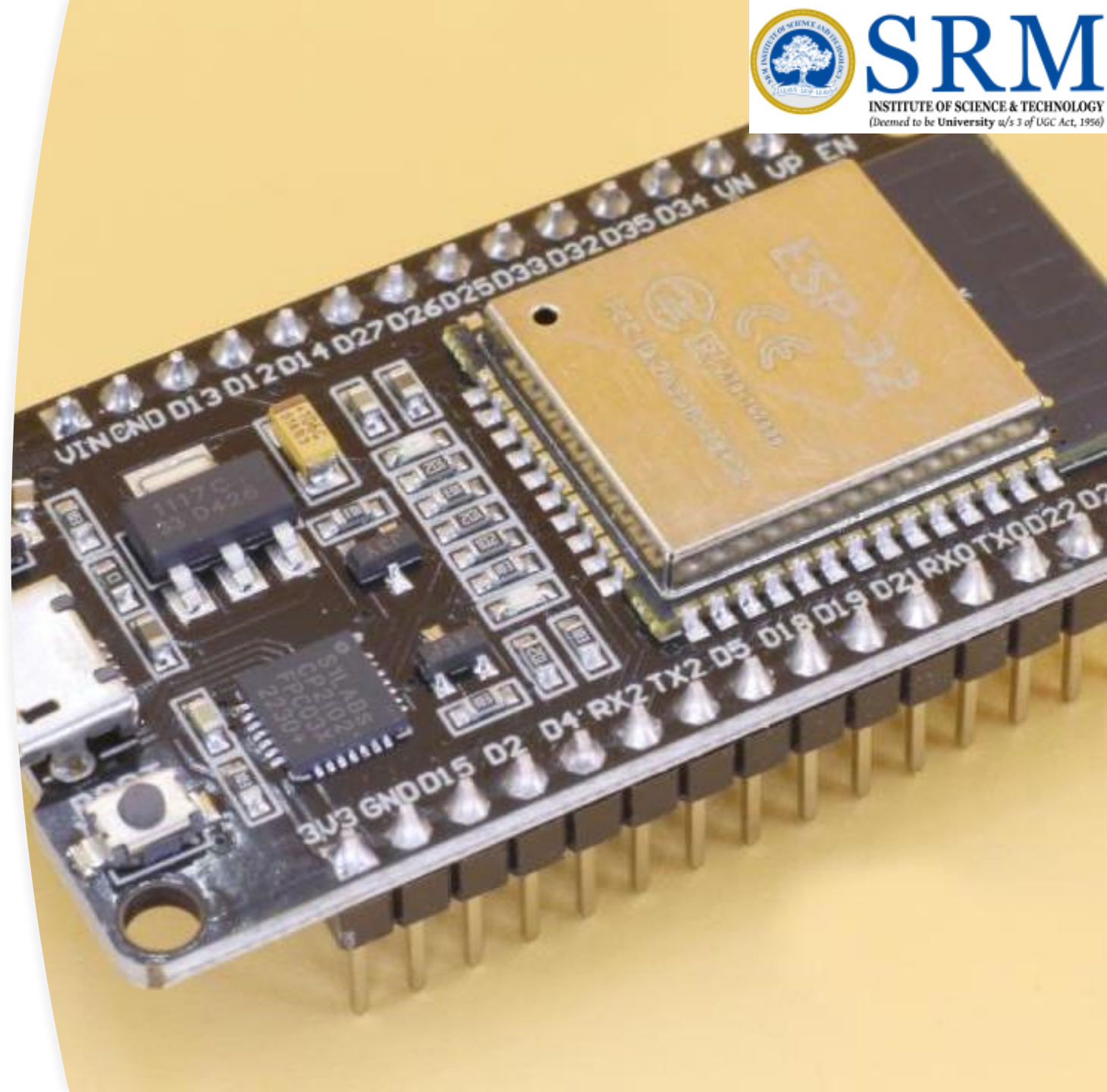
Literature Survey

Sl. No.	Reference	Inference	Link
1	ESP32 Technical Reference Manual, Espressif Systems	Describes ESP32 architecture, Wi-Fi capabilities, and operating modes including Station, AP, and Promiscuous mode used for packet monitoring.	https://www.espressif.com/sites/default/files/documentation/esp32_technical_reference_manual_en.pdf
2	IEEE 802.11 Wireless LAN Standard	Defines Wi-Fi communication protocols, frequency bands (2.4 GHz), and packet transmission structure used in wireless networks.	https://standards.ieee.org/standard/802_11-2020.html
3	Espressif Documentation – Wi-Fi Promiscuous Mode	Explains how ESP32 captures packets without connecting to a network, enabling traffic analysis and packet monitoring.	https://docs.espressif.com/projects/esp-idf/en/latest/esp32/api-guides/wifi.html
4	Spacehuhn ESP8266/ESP32 Deauthentication Project	Demonstrates practical implementation of Wi-Fi deauthentication attack for disconnecting devices from a network.	https://github.com/spacehuhn/esp8266_deauther
5	Stallings, W., Wireless Communications and Networking	Provides theoretical background on wireless communication, network vulnerabilities, and security concepts relevant to Wi-Fi systems.	https://www.pearson.com/en-us/subject-catalog/p/wireless-communications-and-networks/P200000003218



Key Research Findings

- ESP32 supports packet sniffing in promiscuous mode
- Deauthentication attacks can disrupt communication
- Packet monitoring helps analyze network traffic
- OLED interface improves real-time visualization
- Embedded systems enable low-cost implementation



Proposed Methodology

- **Design Flow**
- Study Wi-Fi communication and ESP32
- Setup hardware components
- Implement Wi-Fi scanning
- Enable packet monitoring
- Implement deauthentication-based jamming
- Display results on OLED

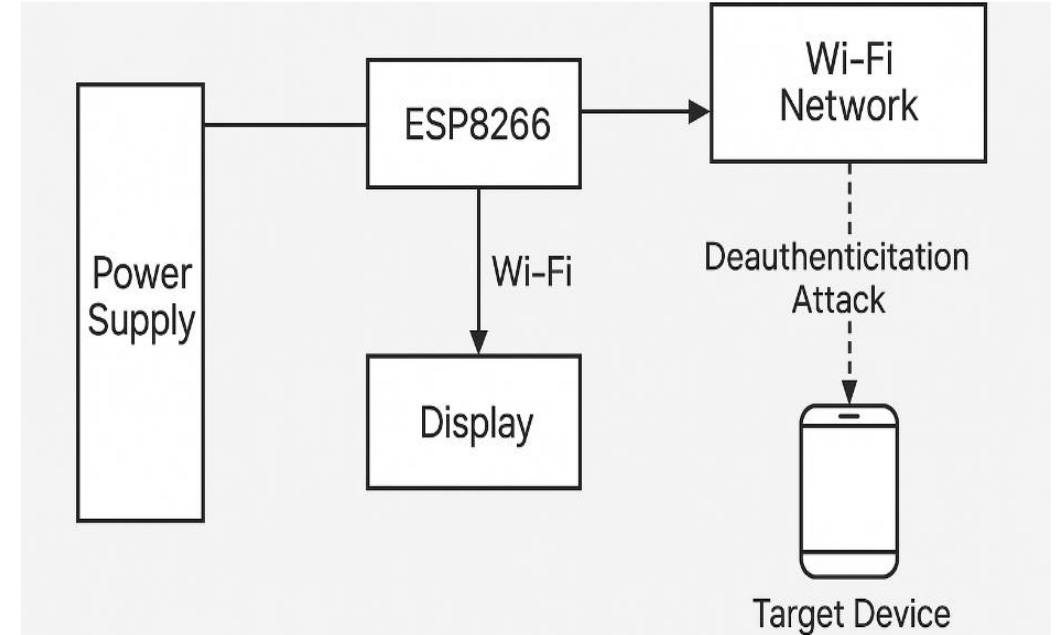


Fig. Block Diagram of Wifi Jammer using ESP 32

Flow Chart :

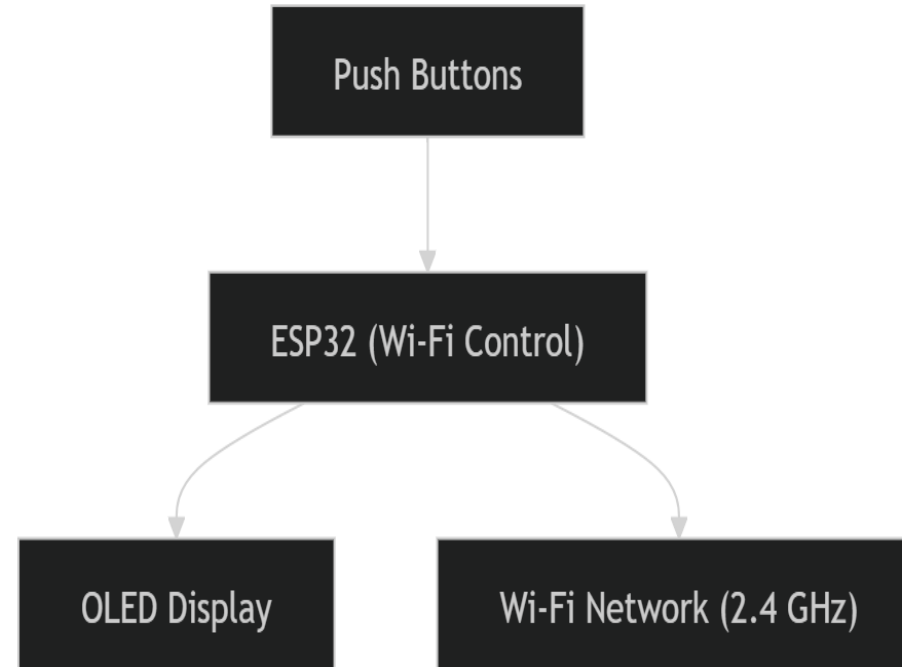


Fig. Flowchart of Wifi Jammer using ESP 32

System Architecture

- **Modules**

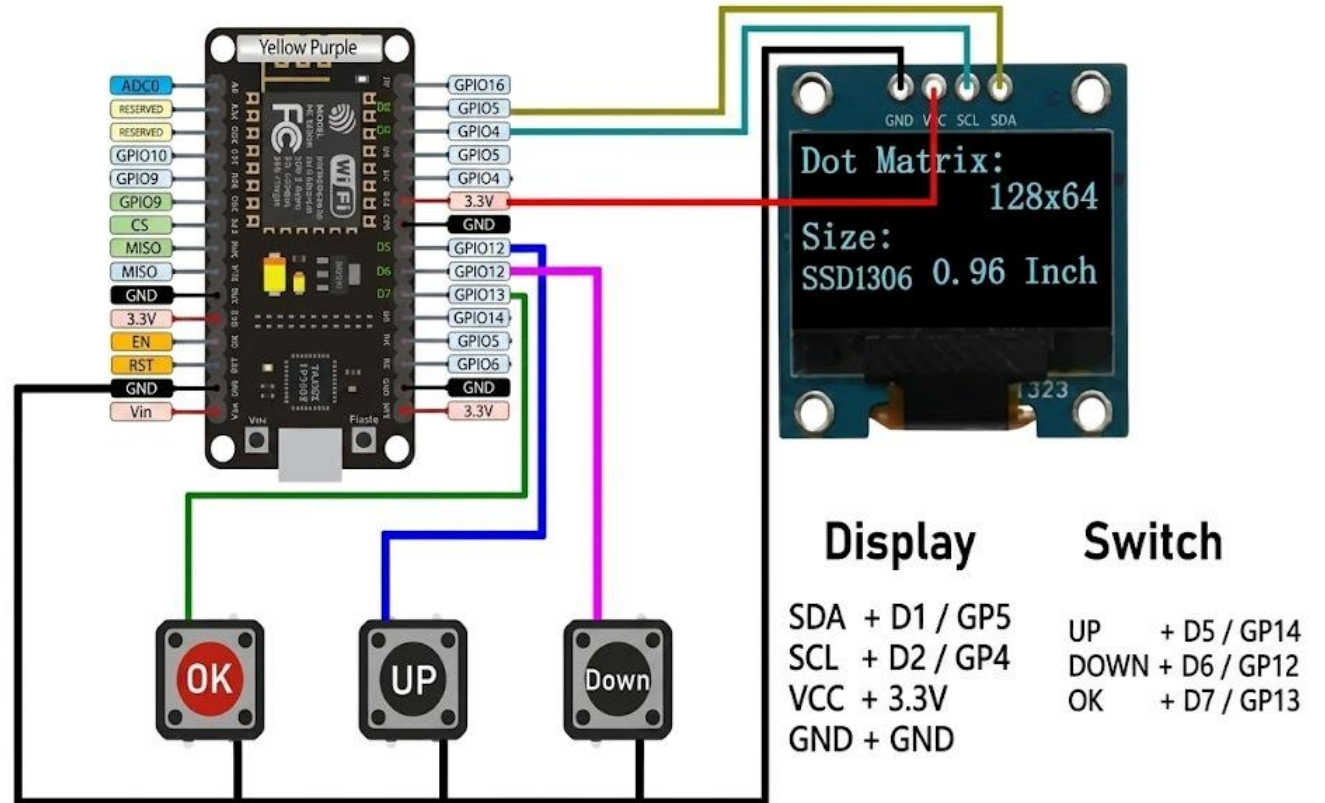
- ESP32 → Main controller
- OLED → Display output
- Push Buttons → User input
- Wi-Fi Network → Communication medium

- **Working**

- Scan → Select → Monitor → Jam

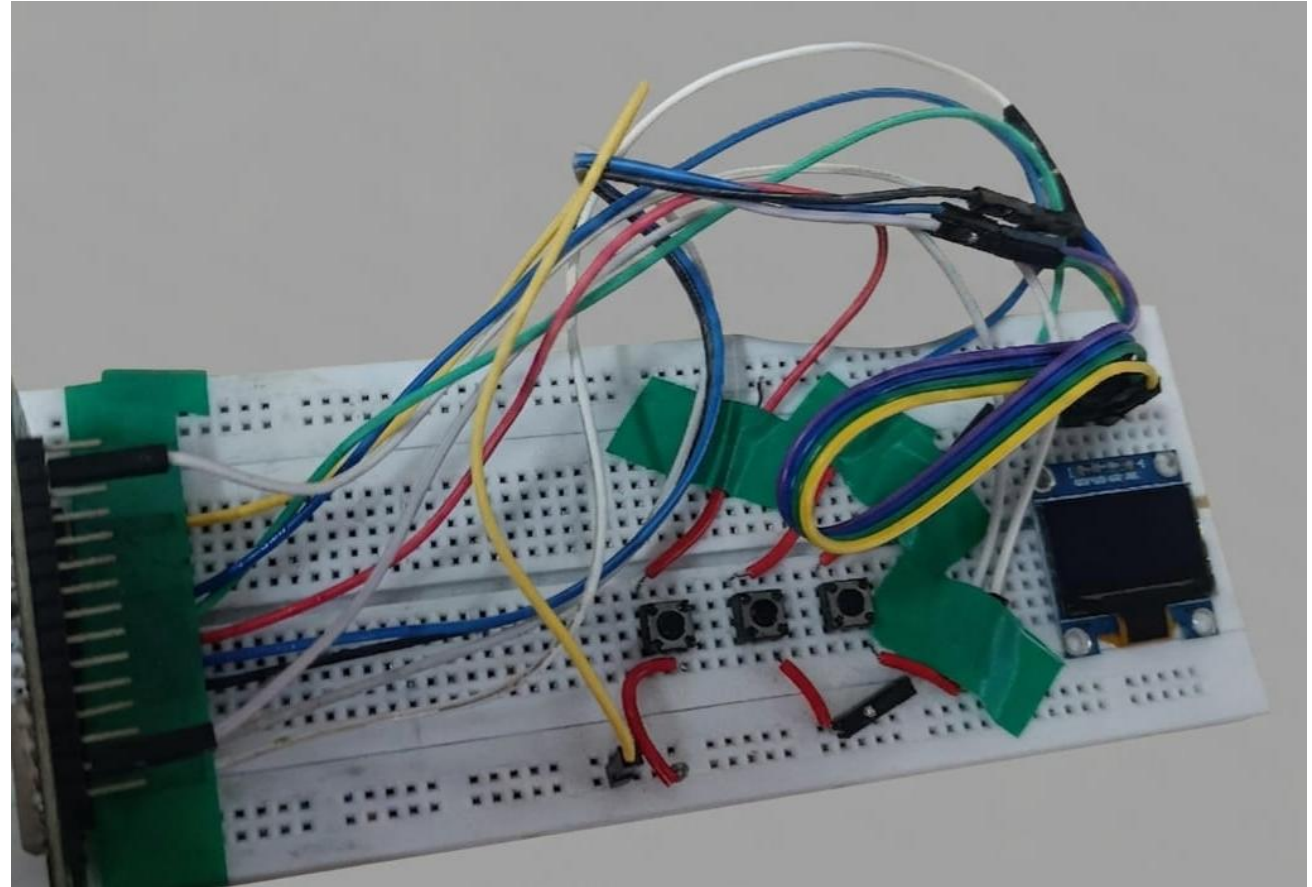
- **Advantages**

- Real-time monitoring
- Portable system
- Low cost



Implementation

- Programming using Arduino IDE
- Libraries used:
 - Wi-Fi library
 - SSD1306 OLED library
- Features implemented:
 - Network scanning
 - Packet counting
 - Button-based control



Expected Results

- Detection of nearby Wi-Fi networks
- Display of SSID, channel
- Packet traffic monitoring
- Successful deauthentication
- Real-time OLED display output

Applications

- Network security testing
- Wi-Fi traffic analysis
- Educational demonstrations
- IoT-based monitoring systems



Conclusion

- Successfully designed Wi-Fi jammer using ESP32
- Demonstrated:
 - Network scanning
 - Packet analysis
 - Deauthentication attack
- Low-cost and efficient system